

DREW ERNST

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Summary

Adept Systems Developer and Data Analyst with a unique background in Public Relations and Marketing, now a self-taught software developer. I specialize in transforming complex data into actionable insights and have a proven track record in architecting systems that prioritize precision and compliance, such as HIPAA-compliant engines and AI-driven finance trackers. I am eager to bring my hands-on architectural approach to a collaborative team focused on large-scale system design and innovation.

Key Achievements

HIPAA Compliant Eliminated server-side vulnerabilities and IT maintenance by architecting a single-file browser application that handles HIPAA-protected information with zero POST requests.

ETL Pipeline Success Engineered full ETL pipeline in Python (Pandas/NumPy) that cleans raw bank CSV data and loads it into a PostgreSQL Star Schema.

Enhanced Financial Tracking Powered a behavioral finance dashboard with real-time net worth tracking, predictive spending analysis, and a 9-step behavioral onboarding flow.

Mobile Optimization Achieved Enabled 100% mobile responsiveness across iOS/Android platforms using Vercel deployment and high-fidelity media queries.

Business and State Consulting Consult on process optimization and data integrity for state agencies and local businesses—bridging technical execution with business strategy.

GitLab & Github Architected and deployed containerized environments using Docker/Nginx with GitHub Actions and GitLab CI/CD pipelines.

Skills

Core Competencies: ETL Pipeline Development • Data Integrity Validation • Behavioral Pattern Recognition • System Architecture Design • API Design & Integration • Component-Based Architecture • CI/CD Pipeline Implementation • Clinical Narrative Validation • SQL Window Functions

Tech Stack: Python (Pandas, NumPy, FastAPI) • React 19, Next.js, Vite • PostgreSQL, SQL, Star Schema Design • TypeScript (Strict Typing) • Docker, Nginx • Git, GitHub Actions, GitLab CI/CD • AWS (S3, EC2) • Jest, Test-Driven Development

Education

Frontend Simplified | 2025 – 2026 *Mentored by David Bragg: Intensive training in zero-latency UI/UX React architecture. Professional-grade engineering standards.*

Quincy University | 2011 – 2012 *Bachelor of Arts in Public Relations*

Culver-Stockton College | 2002 – 2005 *S.I.F.E (Students in Free Enterprise) - Strategic Marketing Lead, developed high-impact outreach strategies and demonstrated entrepreneurial leadership.*

Projects

MCSDD | PCSP Assistant - Missouri State Agency Compliance Engine | 10/2025 <https://dte-solutions.icu/case-study-pcsp/>

- Architected a “Zero-Footprint” Logic Engine that eliminated all server-side privacy risks and hosting fees. By processing HIPAA data locally in volatile RAM, the system ensures 100% data residency with zero POST requests.
- Eliminated Missouri DMH audit rejections by building an automated validation system. Engineered hard logic to enforce state-mandated “Active Treatment” structures and person-centered supports, significantly reducing administrative revision cycles.
- Achieved HIPAA technical compliance without a BAA or database by developing a custom security framework featuring 4-second debounced auto-saves, 30-minute idle timeouts, and 30-day PHI auto-expiry to ensure zero persistent data footprint.

MCSDD | Quarterly Pro - Audit Resilience Engine | 04/2026 <https://dte-84.github.io/case-study-quarterly/>

- Engineered a deterministic middleware that sanitizes PCSP data against the **April 2026 EVV Hard Launch** and **Oct 2025 HRST mandates** to ensure 100% audit finality.
- Directly solved an **80% supervisory oversight gap** identified by state auditors through integrated monthly case review verification and digital timestamping.
- Implemented a Similarity Engine to sanitize clinical narratives against “Rubber Stamping” citations—the root cause of Missouri’s 11% unallowable error rates.

Pulse-Ai: Behavioral Finance Tracker - Full-Stack Data Science | 01/2026 <https://dte-84.github.io/Pulse/>

- Built the entire backend pipeline to handle the messy reality of bank logs—ETL process using Python/Pandas to clean, impute, and prepare raw CSV data for PostgreSQL Star Schema.
- Implemented behavioral logic to flag “Impulse Triggers” by correlating spending velocity with time-of-day and mood indices. Every transformation is validated by Jest child-process tests to ensure mathematical accuracy.
- FastAPI backend with MongoDB/Postgres, utilizing Grouped Imputation techniques for multi-period wealth forecasting.

E-Commerce SQL Analytics - Advanced EDA Repository | 04/2026 <https://ecomm-sql.vercel.app/>

- Developed complex SQL models utilizing **Window Functions (NTILE, DENSE_RANK, LAG)** and CTEs for deep-dive customer RFM segmentation and MoM growth tracking.
- Engineered an automated cohort retention model to track **Lifetime Value (LTV)** and user “stickiness” across multi-stage purchase cycles, identifying a \$1.2M revenue opportunity in under-served clusters.

- Automated data hygiene scripts to prune 2,000+ orphan records and normalize categorical inconsistencies, ensuring 100% integrity for downstream BI reporting.

NestLegacy: Wealth Transfer Intelligence Platform - Behavioral Intelligence | 01/2026 <https://dte-84.github.io/NestLegacy/>

- Designed a digital trust layer to capture “Initial Intent” during the \$84 Trillion Great Wealth Transfer. Identified a cognitive gap where Gen X/Millennial inheritors are 3x more likely to consult AI-driven platforms before engaging human fiduciaries.
 - Built an AI-first lead intelligence system that pre-qualifies users behaviorally, delivering high-fidelity, high-ROI data to CPAs and attorneys.
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Find me online

Github github.com/dte-84

Portfolio dte-84.github.io/DTE-E-Portfolio